Arista Access Node MG-ACC-70 QoS Congestion Event Troubleshooting and Resolution SOP

Node: MG-ACC-70 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_7671

Description

This Standard Operating Procedure (SOP) is designed to mitigate QoS congestion events, characterized by microbursts, interface tail drops, and QoS queue starvation, on the Arista Access Node MG-ACC-70 located at the MG Road Core site. The root cause of these events lies in the node's misconfigured Quality of Service (QoS) policies, which can lead to unfair bandwidth allocation among the various network flows. Additionally, hardware limitations, such as the interface's buffer size and the CPU's processing power, also contribute to the occurrence of these events. Furthermore, the specific Arista CLI dialect in use is designed to facilitate the execution of complex network operations, including QoS policy modifications and interface configuration adjustments. By adhering strictly to the CLI dialect and the troubleshooting methodology outlined in this SOP, network administrators can effectively diagnose and resolve the QoS congestion events on the Arista Access Node MG-ACC-70.

Pre-requisite Checks

- Verify that the remote hands have been dispatched to the MG Road Core site.

- Check the status of the redundant links connected to the Arista Access Node MG-ACC-70.

- Ensure that the network equipment is powered on and functioning within normal operating parameters.

Potential Root Causes

Cause	Probability
Misconfigured QoS policies leading to unfair bandwidth allocation.	High
Hardware limitations, such as interface buffer size and CPU processing power.	Medium
Intermittent network failures or interface flapping.	Low

Diagnostic Commands & Expected Outputs

```
Command: show qos policy-map, Expected Output: Policy Map Configuration: Policy-map-1 Class-map-1: 1000, Class-map-2: 500 Bandwidth: 1000, Priority: High Police: 1000, DSCP: EF , Description: Verify the QoS policy map configuration to identify any misconfigured policies that may be contributing to the congestion event.

Command: show interface counters, Expected Output: Interface GigabitEthernet1/1: 1000 packets transmitted, 900 packets received, 10 errors , Description: Check the interface counters to determine if there are any indications of interface errors or congestion.
```

```
Command: show cpu usage, Expected Output: CPU Usage: 90% ,
Description: Verify the CPU usage to determine if it is within normal
operating parameters or if it is excessively high, indicating a
hardware limitation.
```

Resolution Steps

Step 1: Step: Adjust the QoS policy map configuration, Command: configure terminal policy-map-1 class-map-1 bandwidth 500 , Description: Modify the QoS policy map configuration to adjust the bandwidth allocation for Class-map-1, ensuring that it is fair and within the recommended limits.

Step 2: Step: Reconfigure the interface to increase buffer size, Command: configure terminal interface GigabitEthernet1/1 buffer-size 10000 , Description: Reconfigure the interface to increase the buffer size to prevent interface tail drops and improve network congestion.

Step 3: Step: Verify CPU usage, Command: show cpu usage, Description: Verify that the CPU usage is within normal operating parameters after the resolution steps have been implemented.

Post-Check Verification

```
Command: show qos policy-map, Expected Output: Policy Map
Configuration: Policy-map-1 Class-map-1: 500, Class-map-2: 500
Bandwidth: 500, Priority: Medium Police: 500, DSCP: AF41 ,
Description: Verify that the QoS policy map configuration has been
successfully modified and that the congestion event has been resolved.

Command: show interface counters, Expected Output: Interface
GigabitEthernet1/1: 1000 packets transmitted, 900 packets received, 0
errors , Description: Verify that the interface counters indicate no
errors or congestion after the resolution steps have been implemented.
```

Roll-back Procedures

- Step: Revert the QoS policy map configuration, Command: configure terminal policy-map-1 class-map-1 bandwidth 1000 , Description: Revert the QoS policy map configuration to its original state, restoring the original bandwidth allocation for Class-map-1.

- Step: Revert the interface configuration, Command: configure terminal interface GigabitEthernet1/1 buffer-size 1000 , Description: Revert the interface configuration to its original state, restoring the original buffer size.